# Externalities

Danny Yagan
UC Berkeley
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# Poll (No Right Answer)

The United States should sacrifice immense amounts in order to counter climate change.

- A. True
- B. False
- C. Uncertain

# Outline

- Second part of course is going to cover market failures and show how government interventions can help
  1. Externalities and public goods
  2. Asymmetric information (social insurance)
  3. Individual failures (savings for retirement)

# Externalities

- Market failure: A problem that violates one of the assumptions of the 1st welfare theorem and causes the market economy to deliver an outcome that does not maximize efficiency
- Externality: Externalities arise whenever the actions of one economic agent directly affect another economic agent outside the market mechanism
- Externality example: a steel plant that pollutes a river used for recreation
- Externalities are inefficient and one important case of market failure
- Note: "Pecuniary" externalities redistribute through the price system but are efficient and thus are not real ("technological") externalities. Example: a steel plant uses more electricity and bids up the price of electricity for other electricity customers

# Externality Theory: Economics of Negative Production Externalities

- Negative production externality: When a firm's production reduces the well-being of others who are not compensated by the firm.
- Private marginal cost (PMC): The direct cost to producers of producing an additional unit of a good
- Marginal Damage (MD): Any additional costs associated with the production of the good that are imposed on others but that producers do not pay
- Social marginal cost (SMC $=$ PMC $+$ MD): The private marginal cost to producers plus marginal damage
- Example: steel plant pollutes a river but plant does not face any pollution regulation (and hence ignores pollution when deciding how much to produce)

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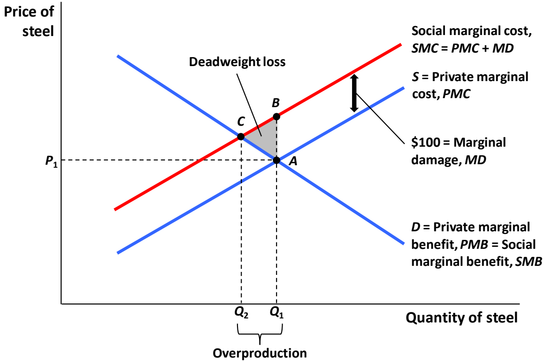
# Economics of Negative Production Externalities: Steel Production



# Externality Theory: Economics of Negative Consumption Externalities

- Negative consumption externality: When an individual's consumption reduces the well-being of others who are not compensated by the individual.
- Private marginal benefit (PMB): The direct benefit to consumers of consuming an additional unit of a good by the consumer.
- Social marginal benefit (SMB): The private marginal benefit to consumers plus any costs associated with the consumption of the good that are imposed on others
- Example: Using a car and emitting carbon contributing to global warming

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## APPLICATION: The Externality of SUVs

The consumption of large cars such as SUVs produces three types of negative externalities:

1. Environmental externalities: Compact cars get 25 miles/gallon, but SUVs get only 20.
2. Wear and tear on roads: Larger cars wear down the roads more.
3. Safety externalities: The odds of having a fatal accident quadruple if the accident is with a typical SUV and not with a car of the same size.

  

# Externality Theory: Positive Externalities

- Positive production externality: When a firm's production increases the well-being of others but the firm is not compensated by those others.
- Example: Beehives of honey producers have a positive impact on pollination and agricultural output
- Positive consumption externality: When an individual's consumption increases the well-being of others but the individual is not compensated by those others.
- Example: Beautiful private garden that passers-by enjoy seeing

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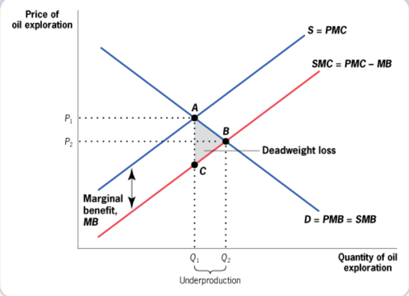
# Externality Theory

## Positive Externalities

■ FIGURE 5-4




**Market Failure Due to Positive Production Externality in the Oil Exploration Market** • Expenditures on oil exploration by any company have a positive externality because they offer more profitable opportunities for other companies. This leads to a social marginal cost that is below the private marginal cost, and a social optimum quantity ($Q_2$) that is greater than the competitive market equilibrium quantity ($Q_1$). There is underproduction of $Q_2 - Q_1$, with an associated deadweight loss of area *ABC*.

# Externality Theory: Market Outcome is Inefficient

- With a free market, quantity and price are such that $PMB = PMC$
- Social optimum is such that $SMB = SMC$
- Negative production externalities lead to over production
- Positive production externalities lead to under production
- Negative consumption externalities lead to over consumption
- Positive consumption externalities lead to under consumption

# Poll

From an efficiency perspective, buildings will tend to have too few murals painted on their sides.

- A. True
- B. False
- C. Uncertain

# Poll

From an efficiency perspective, there are too many households trying to buy homes in the Bay Area.

- A. True
- B. False
- C. Uncertain

# Private-Sector Solutions to Negative Externalities

- Key question raised by Ronald Coase (famous Nobel Prize winner Chicago economist): Are externalities really outside the market mechanism?
- Internalizing the externality: When either private negotiations or government action lead the price to the party to fully reflect the external costs or benefits of that party's actions.

# Private-Sector Solutions to Negative Externalities: Coase Theorem

- Coase Theorem (Part I): When there are well-defined property rights and costless bargaining, then negotiations between the party creating the externality and the party affected by the externality can bring about the socially optimal market quantity.
- Coase Theorem (Part II): The efficient quantity for a good producing an externality does not depend on which party is assigned the property rights, as long as someone is assigned those rights.

# Coase Theorem Example

- Firms pollute a river enjoyed by swimmers. If firms ignore swimmers, there is too much pollution
- Swimmers own river: If river is owned by swimmers then swimmers can charge firms for polluting the river. They will charge firms the marginal damage (MD) per unit of pollution.
    - Why price pollution at MD? If price is above MD, swimmers would want to sell an extra unit of pollution and get hit by pollution damage MD, so price must fall. MD is the equilibrium efficient price in the newly created pollution market.
- Firms own river: If river is owned by firms then firm can charge swimmers in exchange of polluting less. They will also charge swimmers the MD per unit of pollution reduction.
- Final level of pollution will be the same in both cases.

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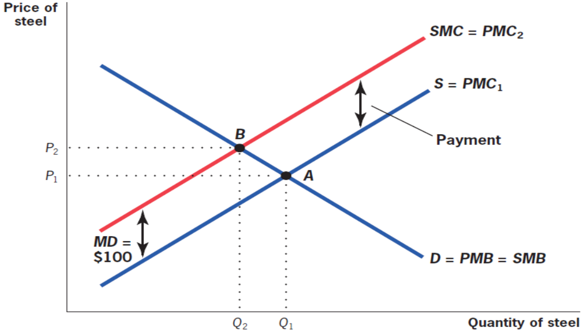
## The Solution: Coasian Payments



# When Does the Coasian Solution Not Work?

- In practice, the Coase theorem solves many but not all types of externalities that cause market failures
- The assignment problem:
  - In cases where externalities affect many agents (e.g. global warming), assigning property rights is difficult.
  - Coasian solutions are likely to be more effective for small, localized externalities than for larger, more global externalities involving large number of people and firms.
- The holdout problem:
  - Shared ownership of property rights gives each owner power over all the others (because joint owners have to all agree to the Coasian solution).
  - As with the assignment problem, the holdout problem would be amplified with an externality involving many parties.

# When Does the Coasian Solution Not Work?

- Transaction Costs and Negotiating Problems:
    - The Coasian approach ignores the fundamental problem that it is hard to negotiate when there are large numbers of individuals on one or both sides of the negotiation.
    - This problem is amplified for an externality such as global warming, where the potentially divergent interests of billions of parties on one side must be somehow aggregated for a negotiation.

# Coasian Solution: Bottom Line

- Coase's insight that externalities can often be internalized was clarifying and powerful
- It means that not every single market failure requires government intervention
- The private market may be able to internalize many small-scale, localized externalities
- Also a reason to suspect that the market will not internalize large-scale global externalities. Possible that only a "government" can aggregate dispersed damages.

# Public Sector Remedies for Externalities

- Public policy makers employ two types of remedies to resolve the problems associated with negative externalities:
- 1) quantity regulation:
  - Government limits use of externality producing chemicals.
  - Example: CFCs [chlorofluorocarbons] that deplete ozone layer.
- 2) corrective taxation:
  - Corrective tax or subsidy equal to marginal damage per unit.
  - Example: carbon tax to fight global warming due to CO2 emissions.
- 1) and 2) can be combined with tradable emissions permits to firms that can then be traded (cap-and-trade for carbon emissions)
- Key advantage of price policy or tradable permits: price of emissions is the same for all which is efficient

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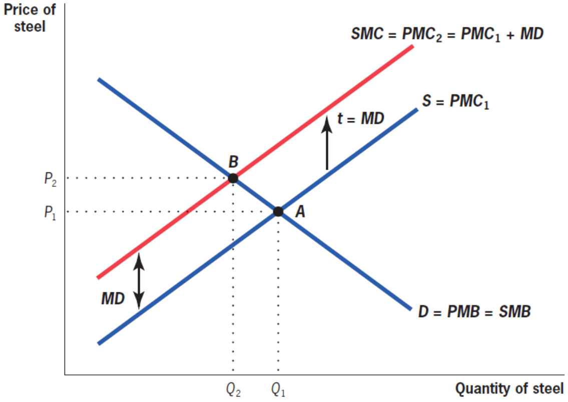
## Corrective Taxation



# Corrective Taxes vs. Tradable Permits

- Two differences between corrective taxes and tradable permits (carbon tax vs. cap-and-trade in the case of CO2 emissions)
- 1) Initial allocation of permits:
  - If the government sells them to firms, this is equivalent to the tax.
  - If the government gives them to current firms for free, this is like the tax + large transfer to initial polluting firms.
- 2) Uncertainty in marginal costs:
  - With uncertainty in costs of reducing pollution, tax cannot target a specific quantity while tradable permits can.
  - Two policies no longer equivalent.
- Taxes preferable when MD curve is flat. Tradable permits are preferable when MD curve is steep.

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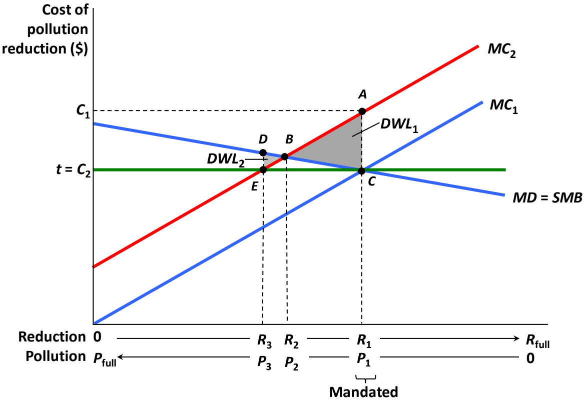
## Uncertainty About Costs of Reduction: Case 1: Flat *MD* Curve (Global Warming)



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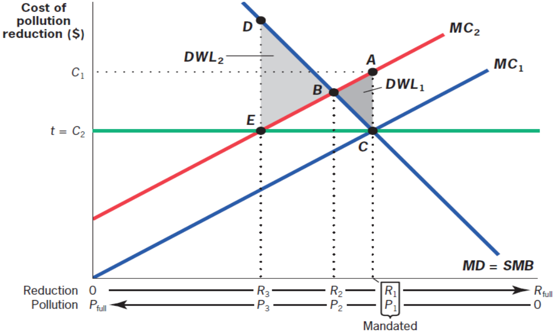
## Uncertainty About Costs of Reduction: Case 2: Steep *MD* Curve (Nuclear leakage)



# Empirical Example: Acid Rain and Health

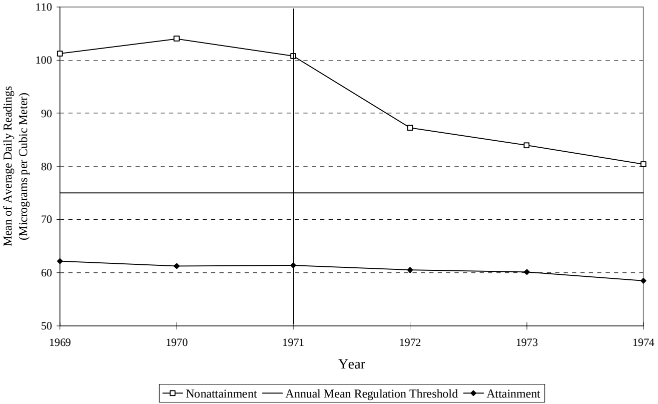
- Acid rain due to contamination by emissions of sulfur dioxide ($SO_2$) and nitrogen oxide ($NO_x$).
- 1970 Clean Air Act:
  - Landmark federal legislation that first regulated acid rain-causing emissions by setting maximum standards for atmospheric concentrations of various substances, including $SO_2$.
- The 1990 Amendments and Emissions Trading:
  - $SO_2$ allowance system: The feature of the 1990 amendments to the Clean Air Act that granted plants permits to emit $SO_2$ in limited quantities and allowed them to trade those permits.

# Empirical Example: Effects of Clean Air Act of 1970

- How does acid rain (or SO2) affect health?
- Observational approach:
  - Relate mortality in a geographical area to the level of particulates (such as SO2) in the air.
- Problem:
  - Areas with more particulates may differ from areas with fewer particulates in many other ways, not just in the amount of particulates in the air.
- Chay and Greenstone (2003) use clean air act of 1970 to resolve the causality problem:
  - Areas with more particulates than threshold required to clean up air [treatment group].
  - Areas with less particulates than threshold are control group.
  - Compares infant mortality across 2 types of places before and after (DD approach).

Figure 2: Trends in TSPs Pollution and Infant Mortality, by 1972 Nonattainment Status

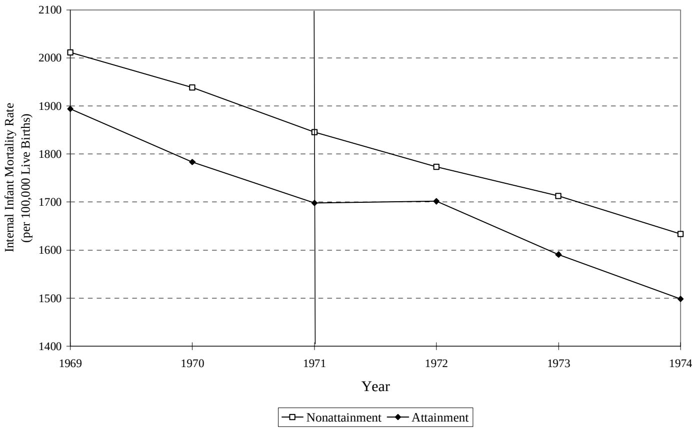
A. Trends in Mean TSPs Concentrations, by 1972 Nonattainment Status




Source: Authors' tabulations from EPA's "Quick Look Reports" data file.

Source: Chay and Greenstone (2003)

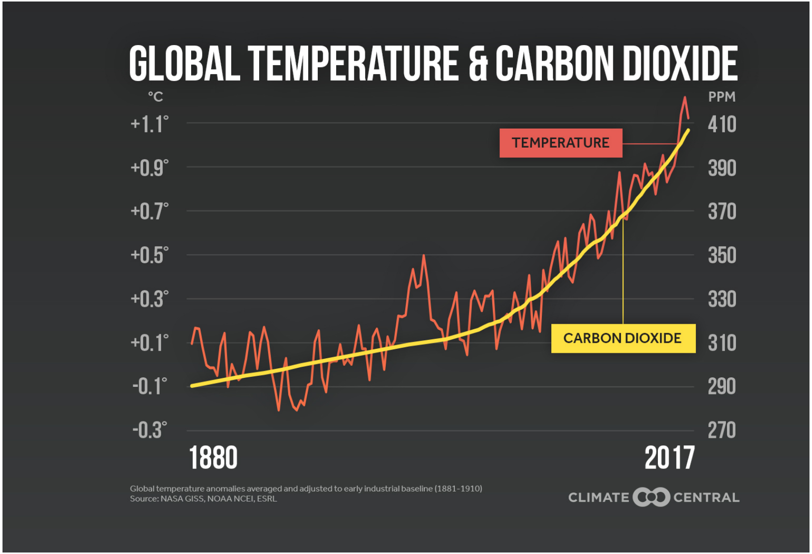
B. Trends in Internal Infant Mortality Rate, by 1972 Nonattainment Status




Source: Chay and Greenstone (2003)

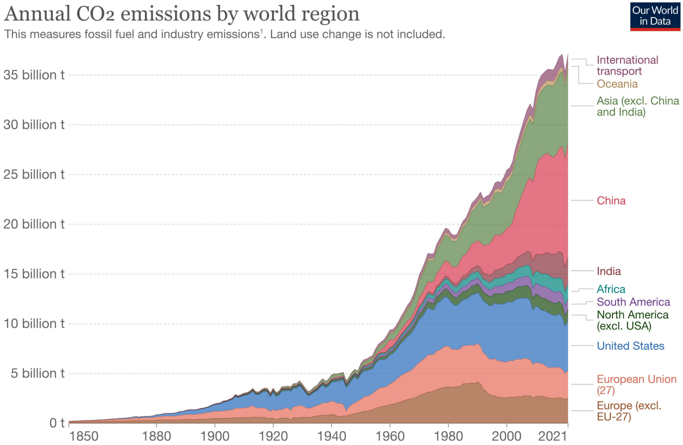
# Climate Change and $CO_2$ Emissions

- Industrialization has dramatically increased $CO_2$ emissions and atmospheric $CO_2$ generates global warming
- Four factors make this challenging (Wagner-Weitzman 2015):
  - Global: Emissions in one country affect the full world.
  - Irreversible: Atmospheric $CO_2$ has long life (centuries) [absent carbon capture tech breakthrough].
  - Long-term: Costs of global warming are decades/centuries away [how should this be discounted?].
  - Uncertain: Great uncertainty in costs of global warming [mitigation vs. amplifying feedback loops].
- How fast should we start reducing emissions? [Stern-Weitzman want a fast reduction, Nordhaus advocates a slower path]


GLOBAL TEMPERATURE & CARBON DIOXIDE
°C
PPM
+1.1°
+0.9°
+0.7°
+0.5°
+0.3°
+0.1°
-0.1°
-0.3°
410
390
370
350
330
310
290
270
TEMPERATURE
CARBON DIOXIDE
1880
2017
Global temperature anomalies averaged and adjusted to early industrial baseline (1881-1910)
Source: NASA GISS, NOAA NCEI, ESRL
CLIMATE CENTRAL

# Annual $CO_2$ emissions by world region

This measures fossil fuel and industry emissions[1]. Land use change is not included.




Source: Our World in Data based on the Global Carbon Project (2022)



**1. Fossil emissions**: Fossil emissions measure the quantity of carbon dioxide ($CO_2$) emitted from the burning of fossil fuels, and directly from industrial processes such as cement and steel production. Fossil $CO_2$ includes emissions from coal, oil, gas, flaring, cement, steel, and other industrial processes. Fossil emissions do not include land use change, deforestation, soils, or vegetation.

# Main Costs of Global Warming

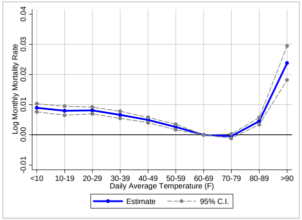
- Enormous variation across geographical areas and economic development. Pace of change makes adaptation daunting
  1. Sea rise will flood low lying coasts and major population centers in many countries (e.g., Miami, Florida; value of real estate subject to regular flooding has dropped)
  2. Impact on bio-diversity (mass extinctions)
  3. Agricultural production could be disrupted by climate change and the increased weather variability it generates: demand for food is very inelastic in the short-run. Spikes in prices if agricultural output falls. Disruption/famines possible in low income countries
  4. Droughts and heat waves will make many places less livable. Some societies may collapse and generate mass migration movements

# Empirical Example: Costs of Global Warming

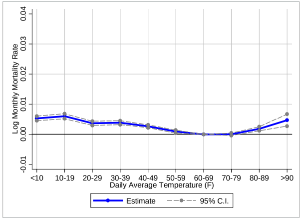
- Estimating costs of Global warming is daunting because society will adapt and reduce costs (relative to a scenario with no adaptation)
- Example: heat waves and mortality analysis of Barreca et al. (2016)
  1. The mortality effect of an extremely hot day (90°F+) declined by about 75% between 1900-1959 and 1960-2004.
  2. Adoption of residential air conditioning (AC) explains the entire decline
  3. Worldwide adoption of AC will speed up the rate of climate change (if fossil fuel powered)

**Figure 2: Estimated Temperature-Mortality Relationship (Continued)**

**(c) 1929-1959**




**(d) 1960-2004**




**Notes:** Figure 2 plots the response function between log monthly mortality rate and average daily temperatures, obtained by fitting Equation (1). The response function is normalized with the 60°F – 69°F category set equal to zero so each estimate corresponds to the estimated impact of an additional day in bin j on the log monthly

**Source:Barreca, Alan, et al (2013)**

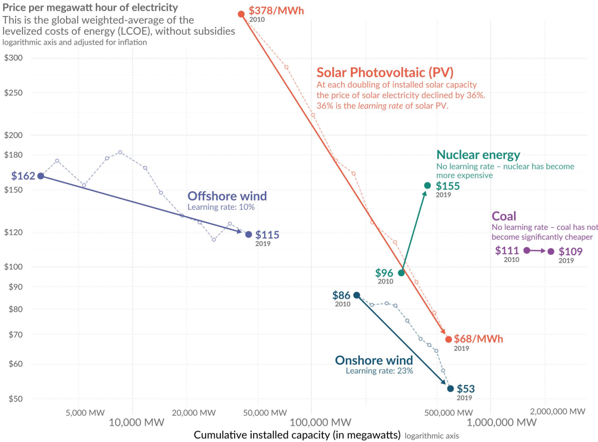
# Global Warming: Economists' Standard View

- Economists view global warming as a classical externality
- CO2 emissions impose a global warming externality. Solution is to impose a carbon tax equal to the marginal damage of CO2 emissions and let market forces work their magic
- Carbon tax proposal often paired with a proposal to rebate the revenue lump-sum (i.e., fixed "carbon dividend" per person)
- But what is the marginal damage of CO2? It depends greatly on how you discount the future
- Economists use interest rate $r$ to discount future: \$1 today is worth $\$(1+r)^T$ in $T$ years (long-distance future heavily discounted: e.g., $r = 4\%$ and $T = 1000 \Rightarrow (1+r)^T = 10^{17}$)
- If interest rate is high, standard economics says it is preferable to let global warming happen

# Global Warming: Alternative View

- Alternative view: if gov directly values future voters (rather than indirectly through current voters), then gov cares more about the future (Farhi-Werning 2007)
- If massive CO2 emissions pose existential civilizational risk (like CFC destroying vital ozone layer), world must decarbonize fast
- Decarbonization: renewable electricity (solar/wind) + grid + big batteries could power most energy needs
- Economists' useful point: some sectors are socially cheaper to decarbonize than others (e.g. cars easier than planes). Start decarbonizing cheaper sectors first (Sachs 2020)
- May require multiple tools (including but not limited to carbon tax) and require major redistribution (to avoid yellow vests protests as in France with higher gas tax)

# Electricity from renewables became cheaper as we increased capacity – electricity from nuclear and coal did not




Source: IRENA 2020 for all data on renewable sources; Lazard for the price of electricity from nuclear and coal – IAEA for nuclear capacity and Global Energy Monitor for coal capacity. Gas is not shown because the price between gas peaker and combined cycles differs signficantly, and global data on the capacity of each of these sources is not available. The price of electricity from gas has fallen over this decade, but over the longer run it is not following a learning curve.

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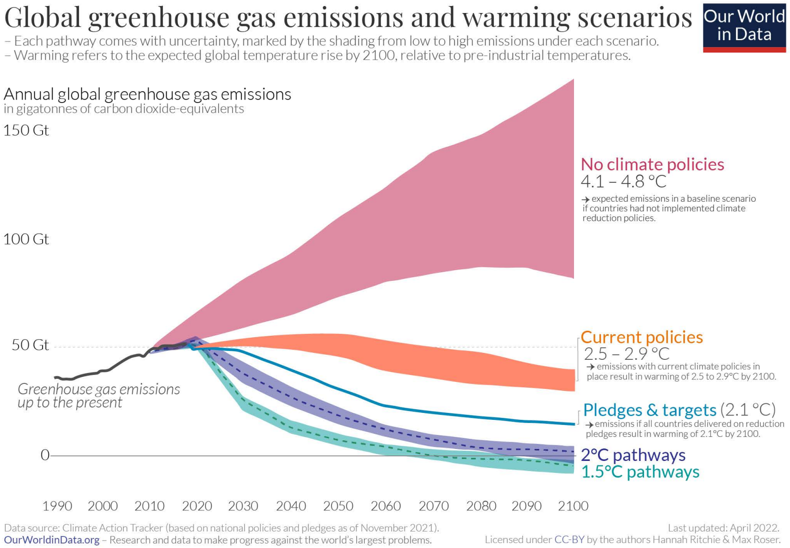
Global greenhouse gas emissions and warming scenarios
Our World in Data
– Each pathway comes with uncertainty, marked by the shading from low to high emissions under each scenario.
– Warming refers to the expected global temperature rise by 2100, relative to pre-industrial temperatures.
Annual global greenhouse gas emissions
in gigatonnes of carbon dioxide-equivalents
150 Gt
100 Gt
50 Gt
0
Greenhouse gas emissions up to the present
No climate policies
4.1 – 4.8 °C
→ expected emissions in a baseline scenario if countries had not implemented climate reduction policies.
Current policies
2.5 – 2.9 °C
→ emissions with current climate policies in place result in warming of 2.5 to 2.9°C by 2100.
Pledges & targets (2.1 °C)
→ emissions if all countries delivered on reduction pledges result in warming of 2.1°C by 2100.
2°C pathways
1.5°C pathways
1990 2000 2010 2020 2030 2040 2050 2060 2070 2080 2090 2100
Data source: Climate Action Tracker (based on national policies and pledges as of November 2021).
OurWorldinData.org – Research and data to make progress against the world's largest problems.
Last updated: April 2022.
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# International Coordination

- From one country perspective, decarbonizing is costly and benefit is modest (as global emissions is what matters)
- Economists: countries need to make a coordinated binding agreement to decarbonize together
- Kyoto 1997: 35 industrialized nations (but not US) agreed to reduce their emissions of greenhouse gases to 1990 levels by 2012 [with ability to trade emission rights among themselves]
- Since then, series of international (but non-binding) pledges
- However, a leader country may have a big impact
- Successful local examples of decarbonization (such as California with its 100% renewable electricity mandate by 2045) may lead to copying

# Richer Countries

- Major policy debates among climate policy advocates
- Encourage research on renewable technologies both public and private (King, David et al. 2015)
- Plan phase out of carbon in various sectors [industrial policy] and weaken fossil fuel industry political power (Sachs 2020)
- Raising carbon tax is economists' preferred tool, but unpopular
- Inflation Reduction Act (2022): Consumer and corporate subsidies, like $7,500 electric vehicle tax credit
- Future: compensate those who lose from climate change policies like a carbon tax?

# Developing Countries

- Disagreement between rich and developing countries on who should bear the cost of curbing greenhouse gas emissions
- Rich countries responsible for most of historical CO2 emissions
- Poor countries want to develop using the cheapest available technologies (coal power is cheaper than solar power, etc.)
- Likely path: richer countries encourage/induce poorer countries leapfrog carbon in favor of renewable energy
- Possible carrot: R&D on renewables in rich countries that can be adopted in poorer countries, possibly with direct subsidies
- Possible stick: Impose tariffs on carbon content of imported goods

# Poll (No Right Answer)

The United States should sacrifice immense amounts in order to counter climate change.

- A. True
- B. False
- C. Uncertain

# Biggest Global Impacts to Date

- International coordination is very hard. Global redistribution is very hard.
- In practice, greatest global gains have come from developed countries subsidizing technological advances that make clean energy privately optimal for the rest of the world
- Example: U.S./German subsidies for solar energy led to technological advances that drive down the cost of solar energy such that China and India voluntarily choose solar farms over fossil fuel power plants for generating electricity
- As of 2026, nine of the top ten largest solar farms in the world are in China and India

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